

Basics: Solutions

Classical Information — Introduction to Quantum Information

1.

- (a) An alphabet of ~ 100 symbols needs enough bit-strings to label all of them; 7 bits give $2^7 = 128 \geq 100$ labels. Equivalently, the number of bits per symbol is $\log(128)/\log(2) = 7$ — a change of base, converting “one 128-symbol character” into “7 binary characters”.
- (b) $500 \times 7 = 3500$ bits.
- (c) Fixed length needs $\lceil \log_2 27 \rceil = 5$ bits per character. $\log_2 27 = 4.755$ bits is the ideal (achieved only asymptotically, by coding long blocks at once). The difference, 0.245 bits per character, is waste forced on us by insisting that every codeword have the same integer length.

2.

- (a) Morse: 16 bits. ASCII: $2 \times 7 = 14$ bits. ASCII wins for this particular pair — Morse is only efficient on text with realistic letter frequencies, and B is a long codeword.
- (b) It is the inter-letter gap. Morse codewords are not prefix-free (e.g. A=10111 begins like E=1), so without a separator 1011100111010101 could be parsed in several ways. The separator is the price paid for a non-prefix-free code.
- (c) Total length is $\sum_x (\text{frequency of } x) \times (\text{length of codeword } x)$. Giving short codewords to high-frequency symbols lowers this average. The optimum, as Shannon showed, assigns roughly $-\log_2 p(x)$ bits to symbol x , giving an average of $H[X]$ bits per character.

3.

- (a) $3000N$ characters.
- (b) ≈ 35 characters for the sentence plus ~ 7 decimal digits for $N = 10^6$: about 42 Latin characters. (In bits: $35 \log_2 27 + \log_2 N \approx 166 + 20 \approx 190$ bits.) The exact count is unimportant — what matters is that it grows only like $\log N$.
- (c) Information per page $\leq (35 + \log N)/N \approx 42/10^6 \approx 4 \times 10^{-5}$ characters per page, $\rightarrow 0$ as $N \rightarrow \infty$. A well-written textbook cannot be summarised this way: its description length grows *proportionally* to N , so its information per page stays finite. Moral: the number of characters on the page is not the information; the length of the shortest description is.

4.

- (a) $H = -0.9 \log_2 0.9 - 0.1 \log_2 0.1 = 0.9(0.152) + 0.1(3.322) = \mathbf{0.469}$ bits.
- (b) $1000 \times 0.469 \approx \mathbf{469}$ bits.
- (c) The raw string is 1000 bits, so we compress to about 47% — a saving of over half, purely because the source is biased.

5.

- (a) $H = -\sum_{x=1}^6 \frac{1}{6} \log_2 \frac{1}{6} = \log_2 6 = \mathbf{2.585}$ bits. (Note this is not an integer: you cannot store one die roll in a whole number of bits without waste, but N rolls cost only $\approx 2.585N$ bits.)
- (b) $H = -\sum_{x=1}^d \frac{1}{d} \log_2 \frac{1}{d} = -d \cdot \frac{1}{d} \log_2 \frac{1}{d} = \log_2 d$.
- (c) Uniform means every outcome is equally likely, so knowing the distribution tells you as little as possible about what will happen: maximal ignorance before the event, maximal information gained on learning it. Any bias gives you a better-than-chance guess, hence less randomness.

6.

- (a) $p = 1 \Rightarrow -\log_2 1 = 0$: a certain outcome carries no surprise and no information. $p \rightarrow 0 \Rightarrow -\log_2 p \rightarrow \infty$: witnessing something you believed impossible is infinitely informative.
- (b) For independent events $p(x_1, x_2) = p(x_1)p(x_2)$, so $-\log p(x_1)p(x_2) = -\log p(x_1) - \log p(x_2)$. The requirement that a function turn products into sums, while being continuous, fixes it to be $\propto \log$; the constant of proportionality is just the choice of units (bits for $\log_2$, nats for $\ln$).
- (c) $\langle S \rangle = \sum_x p(x)(-\log_2 p(x)) = -\sum_x p(x) \log_2 p(x) = H[X]$. So entropy *is* average surprise.
- (d) (i) $-\log_2(1/6) = 2.585$ bits. (ii) $-\log_2(1/2) = 1$ bit — less surprising, as expected.

7.

- (a) $H(0) = 0$; $H(1/4) = H(3/4) = 0.811$; $H(1/2) = 1$; $H(1) = 0$. The sketch is a symmetric dome on $[0, 1]$, zero at both ends, peak 1 at $p = 1/2$, with infinite slope at the endpoints.
- (b) $\frac{dH}{dp} = -\log_2 p - \frac{1}{\ln 2} + \log_2(1-p) + \frac{1}{\ln 2} = \log_2 \frac{1-p}{p}$, which vanishes when $1-p = p$, i.e. $p = 1/2$. Then $H(1/2) = 1$ bit, and $d^2H/dp^2 < 0$ confirms a maximum.
- (c) $H = 0$ means the outcome is certain: N outcomes can be described as “ N zeros”, costing $\log N + \text{const}$ bits in total, i.e. asymptotically *zero bits per outcome*. This is exactly the N -pages-of-zeros example of Q3.

8.

- (a) Average run length $= 1/p(1) = 100$ zeros.
- (b) Recording a number of typical size $1/p(1)$ needs about $\log_2(1/p(1)) = -\log_2 p(1) = 6.64$ bits.
- (c) About $p(1)N$ runs occur, so $K_N \approx -N p(1) \log_2 p(1) = 0.01 \times 6.64 N = 0.066N$ bits, i.e. $H \approx -p(1) \log_2 p(1) = \mathbf{0.066}$ bits per outcome.
- (d) Exact: $H(0.01) = -0.01 \log_2 0.01 - 0.99 \log_2 0.99 = 0.0664 + 0.0144 = \mathbf{0.0808}$ bits. The estimate is *below* the true value because it keeps only the $-p \log p$ term and drops $-(1-p) \log(1-p)$ — the small residual uncertainty carried by the zeros themselves. It is a good approximation precisely because $p(1) \ll 1$.

9.

$$\begin{aligned}
 H[p(y, z)] &= - \sum_{y, z} p(y)p(z) \log_2(p(y)p(z)) \\
 &= - \sum_{y, z} p(y)p(z) \log_2 p(y) - \sum_{y, z} p(y)p(z) \log_2 p(z) \\
 &= - \left(\sum_z p(z) \right) \sum_y p(y) \log_2 p(y) - \left(\sum_y p(y) \right) \sum_z p(z) \log_2 p(z) \\
 &= - \sum_y p(y) \log_2 p(y) - \sum_z p(z) \log_2 p(z) = H[p(y)] + H[p(z)],
 \end{aligned}$$

using $\sum_y p(y) = \sum_z p(z) = 1$. This is condition (3) on any measure of randomness: independent variables contribute independently. Note it also gives $I(Y; Z) = 0$ for independent variables, as it must.

10.

Case	$H(Y)$	$H(Z)$	$H(Y, Z)$	$I(Y; Z)$
(i) independent	1	1	2	0
(ii) perfectly correlated	1	1	1	1
(iii) ★ three equal outcomes	0.918	0.918	1.585	0.252

Case (i) is the direct check of Q9: independence gives $H(Y, Z) = H(Y) + H(Z)$ and hence $I = 0$. Working for (ii): marginals are $p(y=H) = p(y=T) = 1/2$, likewise for z , so $H(Y) = H(Z) = 1$. The joint variable has only two outcomes, each with probability $1/2$, so $H(Y, Z) = 1$. Hence $I = 1 + 1 - 1 = 1$ bit.

For (iii): $p(y=H) = 2/3$, $p(y=T) = 1/3$, so $H(Y) = H(1/3) = 0.918$; same for Z . The joint has three equally likely outcomes, $H(Y, Z) = \log_2 3 = 1.585$. So $I = 2(0.918) - 1.585 = 0.252$ bits.

Interpretation of (ii): each coin individually stores one bit, but it is the *same* bit — the pair holds one bit in total, and the one shared bit is exactly what $I(Y; Z) = 1$ counts.